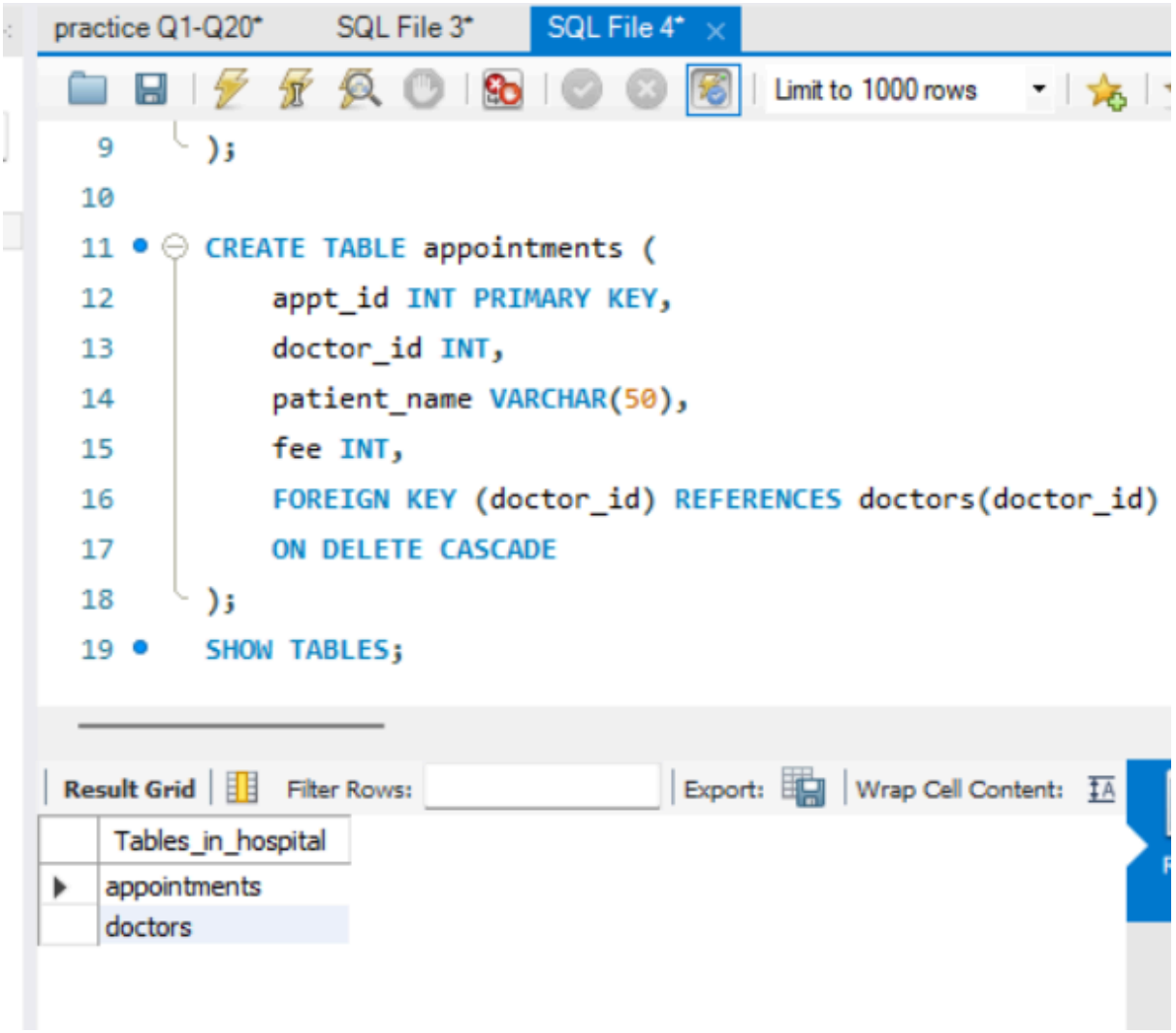
1. Create the hospital database and both tables exactly as shown.

```
CREATE DATABASE hospital;
USE hospital;

CREATE TABLE doctors (
    doctor_id INT PRIMARY KEY,
    doctor_name VARCHAR(50) NOT NULL,
    specialization VARCHAR(50)
);

CREATE TABLE appointments (
    appt_id INT PRIMARY KEY,
    doctor_id INT,
    patient_name VARCHAR(50),
    fee INT,
    FOREIGN KEY (doctor_id)
        REFERENCES doctors(doctor_id)
        ON DELETE CASCADE
);

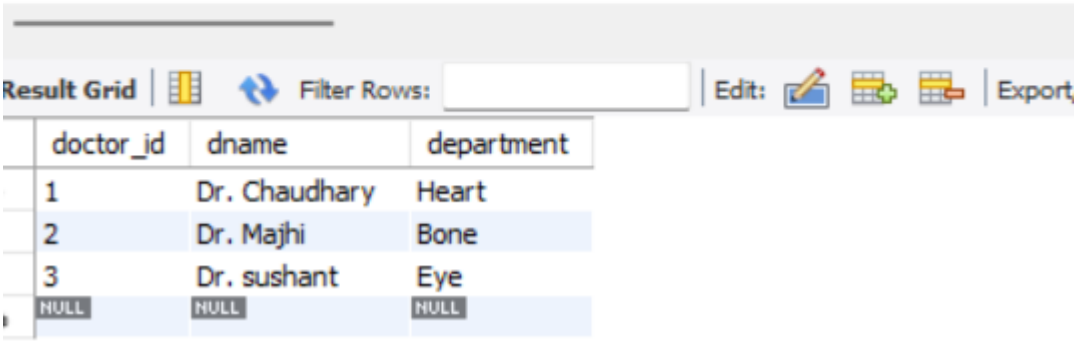
SHOW TABLES;
```



2. Insert 3 doctors: (1, Dr. Chaudhary, Heart), (2, Dr. Majhi, Bone), (3, Dr. sushant, Eye).

INSERT INTO doctors (doctor_id, dname, department)
VALUES
(1, 'Dr. Chaudhary', 'Heart'),
(2, 'Dr. Majhi', 'Bone'),
(3, 'Dr. Sushant', 'Eye');

```
21 •  INSERT INTO doctors (doctor_id, dname, department)
22    VALUES
23    (1, 'Dr. Chaudhary', 'Heart'),
24    (2, 'Dr. Majhi', 'Bone'),
25    (3, 'Dr. sushant', 'Eye');
26
27 •  SELECT * FROM doctors;
```



doctor_id	dname	department
1	Dr. Chaudhary	Heart
2	Dr. Majhi	Bone
3	Dr. sushant	Eye
NULL	NULL	NULL

3. Try to insert a doctor with doctor id = 2 again. Write down the error.

INSERT INTO doctors
VALUES (2, 'Dr. sima', 'Skin');

Error Code: 1062. Duplicate entry '2' for key 'doctors.PRIMARY'

4. Try to insert a doctor with NULL id, and one with NULL name. Which constraint blocks each?

INSERT INTO doctors
VALUES (NULL, 'Dr. Hari', 'Dental');

The `PRIMARY KEY` constraint prevents a NULL ID.

INSERT INTO doctors
VALUES (4, NULL, 'Bone');

The `NOT NULL` constraint on `doctor_name` prevents a NULL name.

5. For each real system, decide a good PK and say why name is not enough: patients in this hospital, products in Daraz, citizens of Nepal.

System	Suitable Primary Key	Why is a name insufficient?
Hospital patients	`patient_id`	Multiple patients can have the same name.
Daraz products	`product_id` or a unique SKU	Different products can have similar names.
Citizens of Nepal	A unique citizenship record identifier, where applicable	Different citizens can share the same name.

A primary key must uniquely identify each record.

6. Insert 4 appointments for existing doctors, two of them for doctor 1. Why is the repeated doctor id allowed?

```
INSERT INTO appointments
VALUES
(1, 1, 'seema', 500),
(2, 1, 'Sushant', 700),
(3, 2, 'sabina', 600),
(4, 3, 'Bhumika', 800);
```

This query inserts four appointment records into the `appointments` table, including appointment IDs, doctor IDs, patient names, and fees. Doctor ID `1` appears twice because one doctor can have multiple appointments.

7. Try to insert an appointment with doctor id = 99. What happens and why is this good?

```
INSERT INTO appointments
VALUES (5, 99, 'Krishna', 500);
```

The query fails with a foreign key constraint error because doctor ID `99` does not exist in the `doctors` table. This is good because it maintains data integrity by preventing appointments from being linked to non-existent doctors.

8. Which table is parent and which is child? Where does the FK live?

- Parent table: `doctors`
- Child table: `appointments`
- Foreign key column: `appointments.doctor_id`

The foreign key is stored in the child table and references the primary key of the parent table.

9. Delete doctor 1. What happened to doctor 1's appointments, and which FK option caused that?

```
DELETE FROM doctors
WHERE doctor_id = 1;
```

Doctor 1 is deleted, and the appointments associated with doctor 1 are also deleted because the foreign key uses `ON DELETE CASCADE`.

10. Recreate the FK with RESTRICT instead. Now try deleting a doctor who has appointments. What happens?

```
ALTER TABLE appointments
DROP FOREIGN KEY appointments_ibfk_1;

ALTER TABLE appointments
ADD CONSTRAINT fk_doctor
FOREIGN KEY (doctor_id)
REFERENCES doctors(doctor_id)
ON DELETE RESTRICT;

DELETE FROM doctors
WHERE doctor_id = 2;
```

`ON DELETE RESTRICT` prevents the deletion of a parent record when child records still reference it.

11. Real life: Daraz has customers and orders. Draw the parent, the child, and the FK column.

Parent table: `customers`

- `customer_id` — Primary Key (PK)
- `customer_name`

One customer can have many orders

Child table: `orders`

- `order_id` — Primary Key (PK)
- `customer_id` — Foreign Key (FK)
- `order_date`

The `orders.customer_id` foreign key references `customers.customer_id`. This relationship allows one customer to have multiple orders.

12. Why can an FK repeat and be NULL while a PK cannot? One sentence each.

Foreign Key: It can repeat because multiple child records may reference the same parent record. It can also be NULL if the column permits NULL values and the relationship is optional.
Primary Key: It cannot repeat or be NULL because it must uniquely identify every row in a table.